

Modeling Informal Paratransit for Local Public Transport Route Planning: A MATSim Application to Tricycle-for-Hire Services in Puerto Princesa City, Philippines

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Abstract:

The lack of formal public transportation in some key cities in the Global South, particularly in Africa and Southeast Asia, has enabled three-wheeled informal public transport vehicles to be popular because of their flexible services, despite sometimes being more expensive than formal public transport. The utter lack of adequate public transport services in such areas has pushed residents residing in such areas to be rather creative in providing local paratransit services usually employing motorcycles retrofitted with a sidecar to service passengers. They are known in different names such as *tricycles* in the Philippines, *tuktuk* in Thailand, *bajaj* in Indonesia, *moto* (motorcycle taxis) in Vietnam and *Keke Naneo* or *Keke Marwa* in Nigeria. These are all responses to the lack of reliable public transport in areas not serviced by higher capacity modes. Common issues associated with three-wheelers, span environmental, safety, operational, regulatory, and socioeconomic dimensions. While they provide essential mobility, their informal nature and technical designs present several persistent challenges.

In the Philippines, tricycle-for-hire services are a common means of transportation. Unlike most land-based public transport modes such as jeepneys and buses that are regulated by the national government, tricycle regulations are issued at the local level (i.e., city or municipal). Local governments issue permits in the form of Motorized Tricycle Operators Permit (MTOP) and regulate fares. Tricycles usually operate as paratransit and have informal operations, having no fixed route nor schedule and operating within a designated area. Being an easy source of employment and due to demands from the community that lack transportation infrastructure and available mobility alternatives, a driver can either own his own tricycle or pay a rental fee to an operator. A driver or an operator is usually a part of a Tricycle Operators and Drivers' Association (TODA). TODAs organize themselves to have different areas of operation and resolve competition. Driver/operators frequently offer "special trips," functioning like a taxi where a passenger pays for the full vehicle capacity to travel directly to a specific destination. For safety and traffic management, they are generally restricted to tertiary or collector roads and are prohibited from plying national highways. Operations are often characterized by high levels of "dead-miles" (unproductive travel without passengers), which can account for 50% to 62% of their total daily distance. In many cities in the Philippines, tricycles narrow roads and residential side streets—where larger vehicles like buses or jeepneys cannot practically operate.

The Public Transport Modernization Program (PTMP), formerly and still commonly referred to as the Public Utility Vehicle Modernization Program (PUVMP) is a program launched by the Department of Transportation (DOTr) in 2017 with the goal of making the Philippines' public transportation system more efficient and environmentally friendly. The program

provides a new regulatory policy where modern public utility vehicles have definite transport routes based on passenger demand and road hierarchy. Vehicles are also consolidated from the current dominance of single ownership towards having a 'common revenue sharing and fleet management' through a 'one-route, one-franchise' system. The so-called Omnibus Franchising Guidelines (OFG), formally known as Department Order No. 2017-011 issued by the Department of Transportation (DOTr), serve as the foundational policy for the Public Utility Vehicle Modernization Program (PUVMP) in the Philippines. At present there are no prescribed Passenger Per Hour Per Direction (*pphph*) thresholds for tricycles within the Omnibus Franchising Guidelines (OFG), which is quite problematic to say the least in terms of local transport planning. This general confusion is demonstrated by more than half of the 1,600 local government units have requested for "Special" Notice of Compliance (NOC) for their Local Public Transport Route Plan (LPTRP) submissions.

There is a need to sufficiently incorporate tricycle-for-hire services into the local public transport route planning as it is driven by regulatory mandates, their critical role in the transport hierarchy, and the necessity to address operational inefficiencies. While tricycles are regulated at the local level, recent national directives have increasingly tied their management to the broader LPTRP framework to ensure a cohesive urban transport system. At present, many tricycle sectors suffer from an over-supply of units, which leads to encroachment on main thoroughfares and increased traffic congestion. Incorporating them into the LPTRP allows for route rationalization studies to determine the appropriate number of vehicles for specific service areas. There is a need to develop proper transport planning tools and modelling framework to analyze tricycle-for-hire operations. Traditional transport modelling, specifically the four-step model, faces significant challenges in representing tricycle-for-hire operations due to its static nature and high level of aggregation. Because tricycles are a form of informal paratransit with non-standard operating patterns, they do not fit the rigid planning assumptions of classical tools.

Traditional models employ spatial and temporal aggregation which constrains the evaluation of tricycle-for-hire services. This makes it nearly impossible to capture the door-to-door, last-mile connectivity that defines tricycle operations. Also, traditional methods rely on aggregation and "averaging demand," which ignores the critical temporal and spatial distribution effects necessary to understand dynamic tricycle requests. A second issue confronting the use of traditional models is the need to analyze the kind of operational informality of tricycles. Traditional models are designed for institutionalized transport with fixed routes, stops, and schedules. Tricycles, however, are unscheduled and ad-hoc, often detouring into narrow streets based on passenger requests. Finally, the "predict and provide" workflow of traditional planning is distinctly mechanistic and struggles to model the complex sensitivities of human behavior. For example, traditional tools cannot anticipate emergent behavior, such as "scavenging" (where tricycles pick up passengers just before a formal bus arrives) or the congestion caused by an over-supply of small-capacity vehicles.

The Multi-Agent Transport Simulation (MATSim) is an activity-based, extendable, multi-agent simulation framework implemented in Java. MATSim uses a microscopic description of demand by tracing the daily schedule and the synthetic travelers' decisions. It offers significant potential for evaluating tricycle-for-hire operations by representing them as a complex, adaptive system where individual agents (passengers) and operators (drivers) interact dynamically. Because tricycles function as an informal paratransit mode providing flexible, point-to-point, and last-mile connectivity, MATSim's core simulation loop is built around fixed-schedule, agent-based mobility, which by itself is insufficient to represent the on-demand, non-scheduled nature of tricycle-for-hire operations. To address this, the framework provides the DVRP contrib, a general-purpose extension for modeling fleets of

vehicles that are dispatched dynamically in response to real-time requests rather than following pre-defined routes. The DRT contrib is built on top of DVRP and provides a ready-made implementation of demand-responsive transport logic—request submission, vehicle assignment, routing, and passenger pickup/drop-off—making it the natural starting point for modeling tricycle behavior, where a passenger effectively hires the full vehicle for a direct, on-demand trip.

The paper presents the development of a MATSim model for evaluating tricycle-for-hire operations taking the case of Puerto Princesa City which is a Highly Urbanized City in the Philippines of about 380,000 residents. Puerto Princesa City serves as the study area given its status as a highly urbanized city with an unusually high dependence on tricycle-for-hire services (48.6% of daily trips) and a road network characterized by low density (0.4 km/km²) due to mountainous terrain in its northern and southern portions. The city's existing policy responses to tricycle oversupply—the 50-50 operating scheme and the more recent zoning scheme—provide a natural set of real-world interventions against which the model's outputs can be compared and offer a basis for scenario design in later stages of the study.

The demand side fuses a locally administered Household Interview Survey (HIS)—rich in behavioral detail but limited in coverage—with the Census of Population and Housing Public Use File (PUF), which offers complete enumeration but lacks trip-behavior data. A synthetic population is generated via Iterative Proportional Updating (IPU), using the `ipfr` R package, to jointly fit household- and person-level marginals from the Census while preserving HIS-observed joint distributions. Data-sparse TAZs use cluster-borrowing from similar neighboring zones to stabilize fitting. Outputs are stored in a SQLite backend feeding both descriptive statistics (e.g., vehicle ownership by barangay/TAZ) and the MATSim population input.

Tricycle-for-hire services are represented in MATSim using a two-mode architecture—**trike_shared** and **trike_special**—corresponding to the two dominant service patterns observed in Philippine tricycle operations: quasi-fixed-route shared rides along habitual corridors, and fully on-demand point-to-point "special trips." Each mode is implemented using a different MATSim mechanism suited to its underlying behavioral logic, operating at two different timescales within the simulation.

- **trike_shared** uses the `minibus` contrib (`org.matsim.contrib.minibus`), based on Neumann's (2014) co-evolutionary paratransit model. Operators evolve routes across iterations via profit-based scoring and mutation-based route modification. The franchise system, stop-provider area restrictions, and activity-weighted demand knowledge map directly onto MTOP franchising, zoning boundaries, and organic hub formation.
- **trike_special** uses the DRT contrib on DVRP, representing on-demand, full-vehicle special trips. Requests are generated per-agent and dispatched within each iteration's mobility simulation via insertion-based assignment, with unserved requests logged as rejected demand. Service-area polygons constrain each fleet to its TODA's franchised zone.

The authors intend to attend the MATSim Use Meeting online and make the presentation remotely as it is quite costly to travel at this time and there is limited time for securing travel approvals including visa application. We are hoping for your kind consideration.

Keywords: Tricycle-for-hire, Demand Responsive, Paratransit, Evolutionary Algorithm, MATSim